

Lab Report No 13
Understanding and practice of LCM display
control to print various messages.
Embedded Systems



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Lab Task No 1:

Solution:

Brief description (3-5 lines)
Write a program which Show the character string “Hello ! Welcome” in the middle of the first line of LCM and blink it every 0.5 second interval.
The code
ORG 0000H ; Start of the code at address 0000H MOV SP, #60 CLR P1.7 CALL LCMINIT MOV A, #10000000B CALL WRINS MOV DPTR, #LINE CALL PRTSTR MAIN_LOOP: CALL PRTSTR CALL DELAY_500MS CLR P1.5 CLR P1.6 CLR P1.7 CALL DELAY_500MS JMP MAIN_LOOP WRINS: CALL CHKBSY CLR P1.5 CLR P1.6 SETB P1.7 MOV P0, A CLR P1.7 RET WRDATA: CALL CHKBSY SETB P1.5 CLR P1.6 SETB P1.7

```

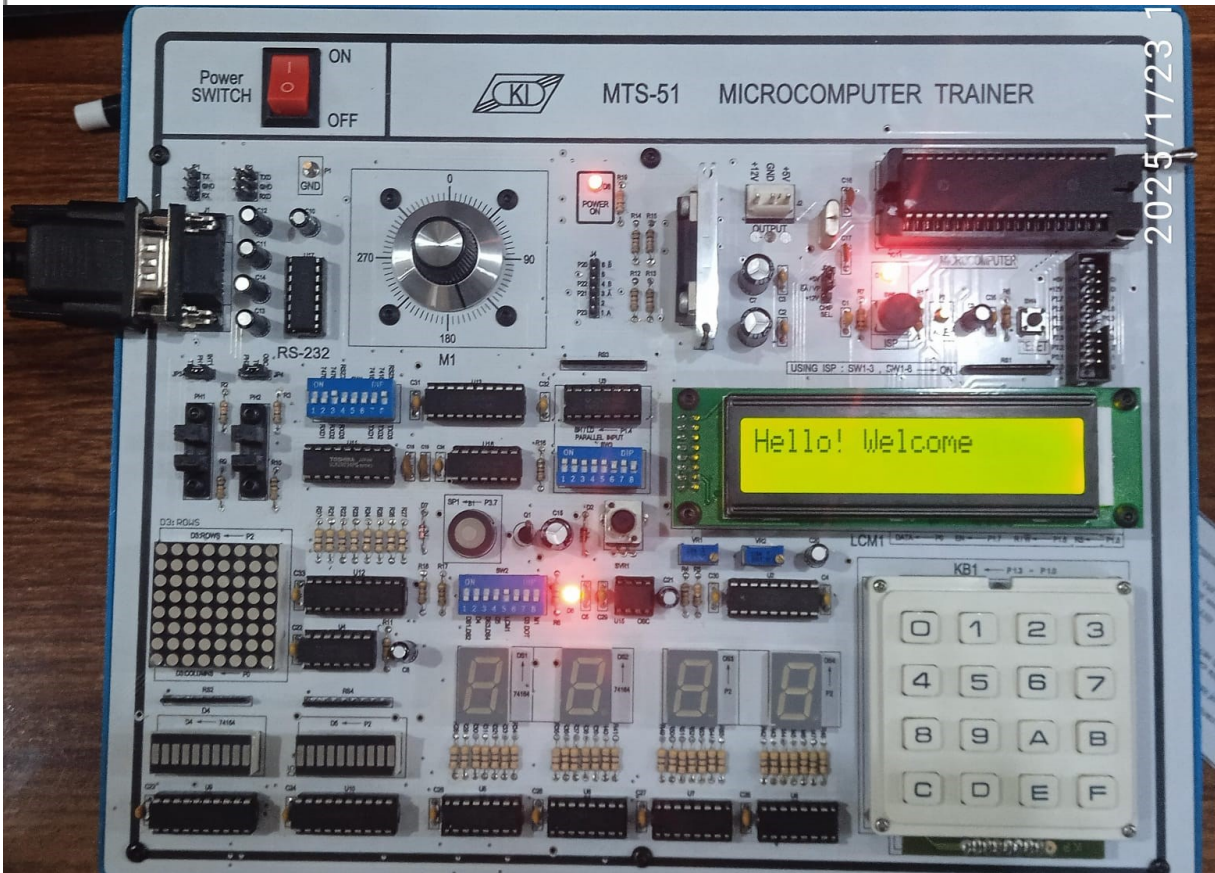
MOV P0, A
CLR P1.7
RET
CHKBSY:
MOV P0, #0FFH
CLR P1.5
SETB P1.6
SETB P1.7
MOV C, P0.7
CLR P1.7
JC CHKBSY
RET
LCMINIT:
MOV A, #00110000B
CALL WRINS
MOV A, #00000001B
CALL WRINS
MOV A, #00001100B
CALL WRINS
MOV A, #00000110B
CALL WRINS
RET
PRTSTR:
MOV A, #0
MOVC A, @A + DPTR
CJNE A, #'$', PRINT
RET
PRINT:
CALL WRDATA
INC DPTR
JMP PRTSTR
DELAY_500MS:
MOV R6, #50
DELAY_LOOP_500MS:
MOV R7, #200
DELAY_LOOP_200MS:
DJNZ R7, DELAY_LOOP_200MS
DJNZ R6, DELAY_LOOP_500MS
RET
LINE: DB 'Hello! Welcome$'

```

end

The results (Screenshot)

```
Program Size: data=9.0 xdata=0 code=105  
creating hex file from ".\Objects\LCM"...  
".\Objects\LCM" - 0 Error(s), 3 Warning(s).  
Build Time Elapsed: 00:00:00
```



Conclusion

In conclusion, we had performed the LCM display control to print various messages. In the assembly language to print hello and welcome string in the Keil 8051 and its show in flash magic.